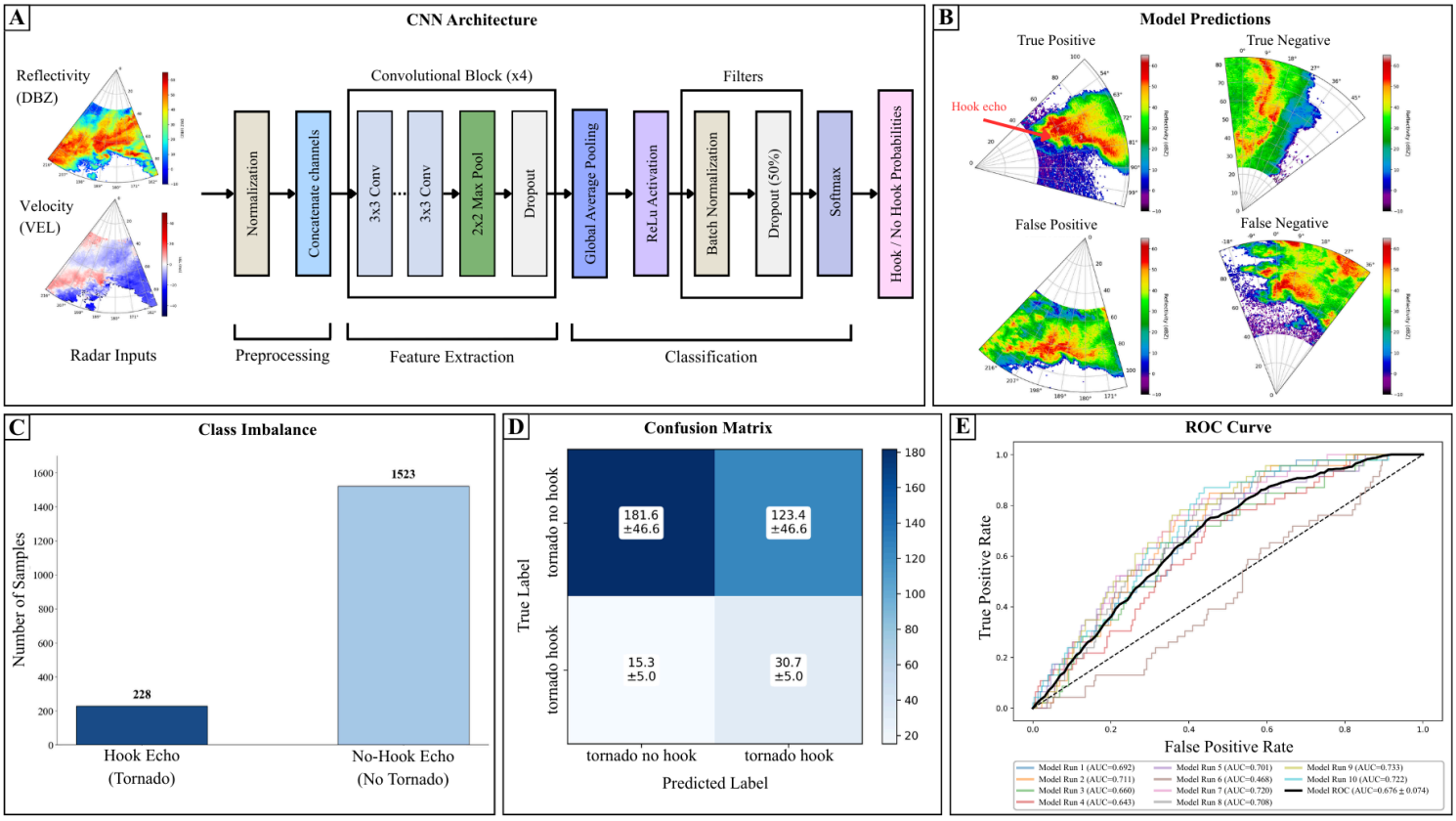

Figure 1. CNN-based classification of tornado hook echoes from Doppler radar data. **(A)** Model architecture showing how dual-channel radar inputs (reflectivity DBZ and velocity VEL) are normalized, concatenated, and processed through four convolutional blocks before being classified as hook or no-hook. **(B)** Example radar reflectivity scans for each prediction outcome including True Positive (hook correctly identified, red arrow), True Negative, False Positive, and False Negative. **(C)** Dataset class distribution showing 228 hook echo samples versus 1,523 no-hook samples, a roughly 7:1 ratio that was addressed during training using balanced batching and data augmentation. **(D)** Mean confusion matrix averaged across 10 model runs showing counts with standard deviation, where the model performs better on the majority class than on hook detection. **(E)** Mean ROC curve across 10 runs with an AUC of 0.676 ± 0.074, with each individual run shown in a different color.